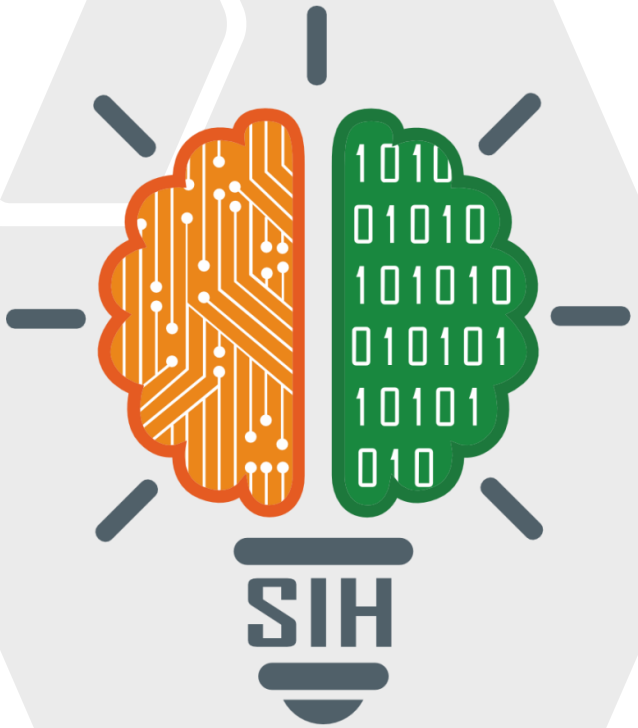
SMART INDIA HACKATHON 2026

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- **Problem Statement ID –** 26047
- **Problem Statement Title –** Patient Case-Taking Software
- **Proposed Solution –** MediKiosk: AI history-taking by phone call, in the patient's own language
- **Organisation –** Ministry of Ayush · All India Institute of Ayurveda
- **Theme –** MedTech / BioTech / HealthTech
- **PS Category –** Software
- **Team ID –**
- **Team Name (Registered on portal) –** Taraket



Taraket

MediKiosk: The History Is Taken Before the Patient Walks In

SMART INDIA HACKATHON 2026

Real-world issue: An OPD doctor gets **2–5 min** per patient for history, exam, old papers and prescription.

Why important: A well-taken history alone gives the right diagnosis in **70-80 %** of cases.

Our solution: An AI phone call takes the full history in the patient's language. The doctor reads it in **30 s**.

ANY PHONE · ANY INDIAN LANGUAGE

हिन्दी · తెలుగు · English · தமிழ் · বাংলা · मराठी
ಕನ್ನಡ · മലയാളം · ગુજરાતી · ଓଡ଼ିଆ · ਪੰਜਾਬੀ

No app. No typing. No reading.

THE COMPLETE LOOP · CALL IN, SUMMARY OUT

PATIENT speaks
दो दिन से सीने में दर्द है

↓

AI INTERVIEW
asks like a doctor: onset, spread, severity…

↓

STRUCTURED HISTORY IN SECONDS
CC · HPI · Past · Drugs · Family · ROS · Ayurveda

↓

OLD PAPERS DIGITISED
prescriptions & lab reports → one timeline

↓

DOCTOR confirms
edits, signs off, sends to HIS + ABHA

TODAY	→	WITH MEDIKIOSK
2–5 min for everything		**30 s** to read the full history

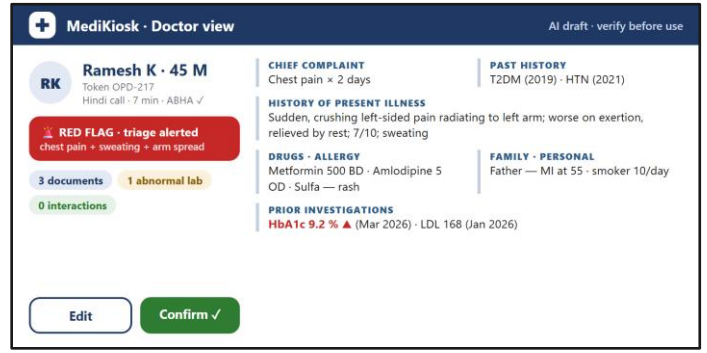
History read, not asked · the consultation goes back to care

PROBLEM v/s MEDIKIOSK

PROBLEM		MEDIKIOSK
No time to ask	⇔	AI asks before the visit
Apps shut out elders	⇔	Works on any phone
A bag of old papers	⇔	Read into a timeline

EMERGENCY

Chest pain in the queue?	⇔	Triage alerted in seconds



Any phone. Any language. No app.

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TECHNICAL APPROACH

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METHODOLOGY & PROCESS

1. **IDENTIFY**
   ABHA / mobile OTP · spoken consent (DPDP)
2. **CONVERSE**
   voice + keypad interview, ontology-driven
3. **SAFEGUARD**
   red-flag rules run after every answer
4. **DIGITISE**
   OCR → medicines, labs, dates → timeline
5. **SUMMARISE**
   fact-checked draft · every line sourced
6. **CONFIRM**
   doctor edits → FHIR to HIS + ABHA

[Detailed Report](#) [GitHub](#)

THE APPLICATION



Every answer is read back: the patient confirms, not the AI.

TECHNOLOGIES USED

VOICE CALL	Pipecat · Silero VAD · Exotel / WebRTC · keypad (DTMF) fallback
SPEECH AI	Sarvam Saaras v3 STT, handles 8 kHz phone audio · Bulbul v3 TTS
CLINICAL BRAIN	SOCRATES question ontology + Sarvam-30B; Sarvam-105B writes the summary
DOCUMENT AI	Sarvam Vision OCR · Claude Haiku 4.5 for hard handwriting
ABDM & DATA	FHIR R4 · ABHA · SNOMED CT · LOINC · NAMASTE · FastAPI · PostgreSQL

SAFETY — IT CANNOT INVENT A HISTORY

ONTOLOGY ASKS
the LLM only reads answers; it never invents a question

RULES ESCALATE
chest pain + breathlessness → nurse alerted at once

AI drafts. The doctor decides.

Every summary line links to a patient answer or a document: no source, no sentence.

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FEASIBILITY AND VIABILITY

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THE CONSULTATION HAS NO TIME FOR HISTORY — so MediKiosk takes it before the patient walks in

TODAY	HISTORY · asked in a rush, often incomplete	EXAMINATION	PRESCRIPTION
WITH MEDIKIOSK	READ 30 s	EXAMINATION · REASONING · COUNSELLING · PRESCRIPTION	

the same 2–5 minute consultation · illustrative split

HISTORY MOVES OUT OF THE ROOM
taken on a phone call while the patient waits

Same doctor. Same minutes.
Minutes spent asking become minutes spent examining and explaining.

FEASIBILITY — CAN WE BUILD IT?

Nothing to invent
every AI block exists as an API or open model

Runs on what India has
any phone; kiosk = touchscreen + handset

Standards ready
ABDM sandbox, FHIR R4, SNOMED CT

cloud APIs **₹5–10** → on-premise **≈ ₹0**

AI cost per patient · estimates; on-prem excludes hardware

VIABILITY — WILL IT LAST?

Deploys in any OPD
no new staff, no patient training

Scales by configuration
a new language = translated questions + audio

Indian and open
Sarvam models (IndiaAI Mission), open weights

Built for national policy
ABDM · DPDP Act 2023 · NAMASTE codes

RISKS — AND THE ANSWER

Accents & noisy OPDs
→ phone-audio STT, read-back, keypad fallback

AI hallucination
→ ontology asks · fact-check · doctor confirms

A missed emergency
→ deterministic rules, 100 % test suite

Health-data privacy
→ consent-first, encrypted, on-prem option

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IMPACT AND BENEFITS

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WHO IT REACHES

Rural patients
heard in their own language

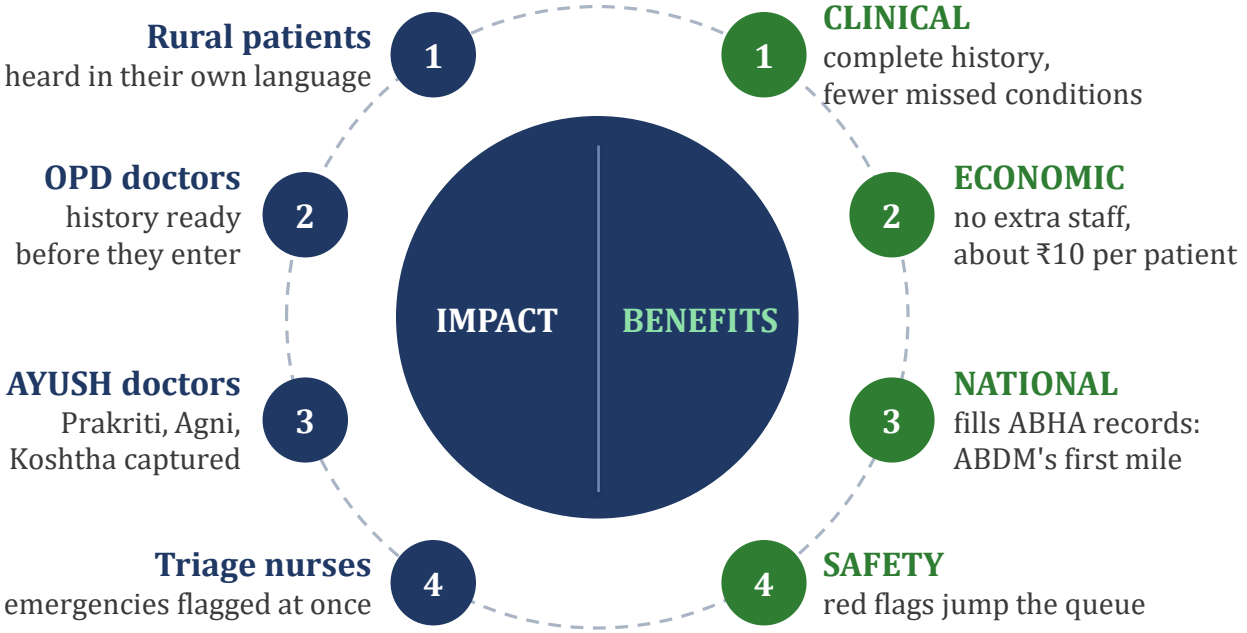
OPD doctors
history ready before they enter

AYUSH doctors
Prakriti, Agni, Koshtha captured

Triage nurses
emergencies flagged at once

WHAT CHANGES

Any phone. Any language. No app, no typing, no training.



A grandmother speaking Telugu, a farmer with a bag of old papers. **One call, and the doctor already knows.**

THE PROBLEM, IN NUMBERS

up to
10,000
OPD patients a day in one tertiary hospital

2–5 min
for history, exam and prescription

70-80 %
of diagnoses reachable from history alone

Figures from the PS 26047 background, incl. a BMJ Open (2017) study of consultation length across 67 countries.

TARGETS WE WILL PROVE

100 % **SAFETY**
red-flag test suite in 3 languages · zero missed emergencies

≤ 1.5 s **SPEED**
per voice turn · doctor reads the summary in under 30 s

8 / 8 **COMPLETENESS**
history sections for every patient, plus Ayurveda

Our evaluation suite reports these numbers: **checkable, not claimed.**

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RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

the problem

BMJ Open 2017

67-country study

LINK

the health stack

ABDM Sandbox

ABHA · consent · HIP

LINK

the record format

NRCeS FHIR R4 IG

OPConsultRecord

LINK

the privacy law

DPDP Act 2023

consent-first design

LINK

hears and speaks

Sarvam AI

Saaras · Bulbul · 30B

LINK

runs in-house

AI4Bharat

open speech models

LINK

Ayurveda codes

NAMASTE & ICD-11 TM2

Ayush · WHO

LINK



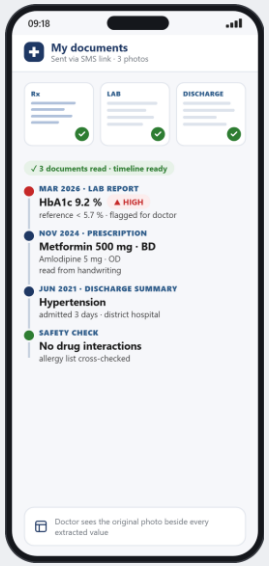
Calls any phone



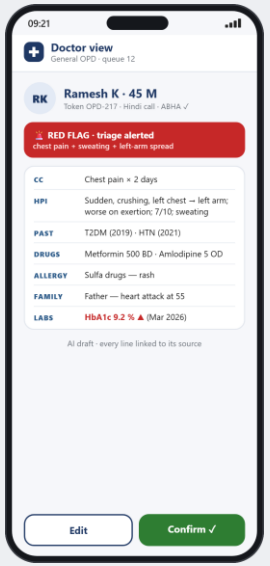
Own language, always



Speak or tap



Papers → timeline



Doctor reads in 30 s

UI / UX

COMPLIANCE

DPDP Act 2023: spoken consent, per purpose, revocable. ABDM: ABHA-linked FHIR R4 records. The AI only drafts and never diagnoses; a doctor confirms every summary. Runs fully on-premise if required.

Detailed Report

GitHub

13 project documents: architecture, API, clinical design, security and test plan.